

Beyond Importance Sampling: Reducing Variance for Free

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$SE(\hat{\theta}_n) = \sigma / \sqrt{n}$: the only way to improve precision *without* more computation is to reduce $\sigma^2 = \text{Var}[h(X)]$ (or its importance-sampling analogue).

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2. **Control variates** – exploit a correlated quantity whose expectation is already known.

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$$\Rightarrow \underline{\hat{\theta}_n \pm z_{\alpha/2} \sigma / \sqrt{n}}$$

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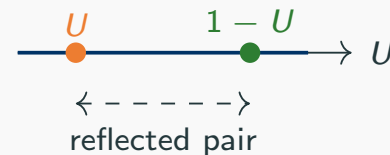
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1. **Antithetic variates** – introduce negative correlation between paired draws.
2. **Control variates** – exploit a correlated quantity whose expectation is already known.
3. **Stratified sampling** – force samples to spread evenly across the domain.

Antithetic Variates: Idea and Algorithm

$$\Rightarrow h_1(U) \rightarrow h_2(U)$$

- For $\underline{U \sim \text{Unif}(0, 1)}$, note $\underline{1 - U \sim \text{Unif}(0, 1)}$ too, and $U, 1 - U$ are **negatively correlated**.



$$U \sim \text{Unif}[0, 1]$$

$$X = 1 - U \sim \text{Unif}[0, 1] \Rightarrow F_X(x) = x$$

$$F_X(x) = P(X \leq x) = P(1 - U \leq x) = P(1 - x \leq U)$$

$$= P(U \geq 1 - x)$$

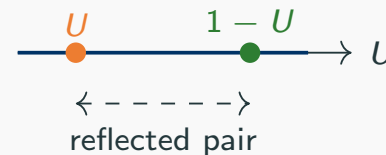
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$\sim U[0, 1]$

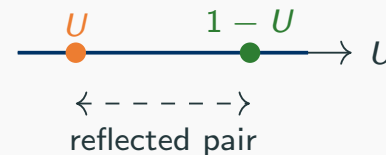
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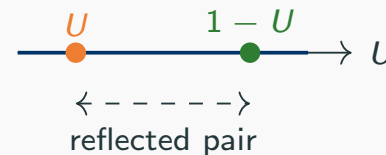
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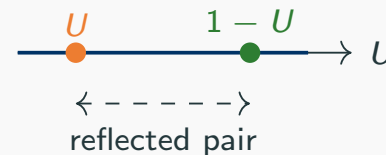
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Algorithm: Antithetic Variates

1. Draw $U_1, \dots, U_{n/2} \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$. ✓
2. Form the n paired values $h(U_i)$ and $h(1 - U_i)$ for each i .
3. Return $\hat{\theta}_n^{\text{AV}} = \frac{1}{n} \sum_{i=1}^{n/2} [h(U_i) + h(1 - U_i)]$. $\Rightarrow$

$$\theta_h^{\text{CMC}} = \frac{1}{n} \sum_{i=1}^n h(x_i)$$
$$\rightarrow V(\theta_n^{\text{AV}}) = \frac{\text{Var}(h(U))}{n}$$

$$V(h(U)) = \text{Var}(h(U) + h(1 - U)) =$$

Antithetic Variates: Why It Works

Variance comparison

Let $\underline{Y_i} = \frac{h(U_i) + h(1 - U_i)}{2}$. Then

$$\underline{\text{Var}(Y_i)} = \frac{1}{4} \left[\underline{\text{Var}(h(U))} + \underline{\text{Var}(h(1 - U))} + 2 \underline{\text{Cov}(h(U), h(1 - U))} \right].$$

If $\underline{h}$ is monotonic, $\underline{\text{Cov}(h(U), h(1 - U))} < 0$, so pairing **strictly reduces** variance compared to $\underline{n/2}$ independent pairs of ordinary crude draws.

↑
crude mc

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- **Finance application:** simulate an option payoff on both a price path driven by Z_i and the “mirror” path driven by $-Z_i$ – a standard variance-reduction trick for Asian and other path-dependent options.
- Cheap to implement (no extra function evaluations needed if we already need n total draws) but the benefit depends on h being reasonably monotonic.

Control Variates: Idea and Algorithm

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- Define the **controlled estimator**

$$h_\beta(X) = h(X) - \beta(c(X) - \mu_c).$$

$$\begin{aligned} E_f(h_\beta(X)) &= E_f[h(X) - \beta(c(X) - \mu_c)] \\ &= E_f[h(X)] - \beta E_f[c(X) - \mu_c] \\ &= E_f[h(X)] \end{aligned}$$

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- Since $E[c(X) - \mu_c] = 0$, we still have $E[h_\beta(X)] = \theta$ for any β – unbiased for every choice.

⇒ $\beta^* = \text{Cov}(h(X), c(X)) / \text{Var}(c(X))$ is optimal

Use $\text{Var}(h - \beta(c - \mu_c)) = \text{Var}(h) + \beta^2 \text{Var}(c) - 2\beta \text{Cov}(h, c)$. This is a quadratic in β ; differentiate and set to zero: $2\beta \text{Var}(c) - 2\text{Cov}(h, c) = 0 \Rightarrow \beta^* = \text{Cov}(h, c) / \text{Var}(c)$, giving the minimum variance.

$$V(X+Y) = V(X) + V(Y) + 2\text{Cov}(X, Y)$$

$$V(X-Y) = V(X) + V(Y) - 2\text{Cov}(X, Y)$$

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Algorithm: Control Variates

1. Draw $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$; estimate $\hat{\beta} = \widehat{\text{Cov}}(h(X), c(X)) / \widehat{\text{Var}}(c(X))$ (optimal choice).
2. Return $\hat{\theta}_n^{\text{CV}} = \frac{1}{n} \sum_{i=1}^n [h(X_i) - \hat{\beta}(c(X_i) - \mu_c)]$.

Control Variates: Variance Reduction and an Example

Optimal variance reduction

With $\beta^* = \text{Cov}(h(X), c(X)) / \text{Var}(c(X))$,

$$h_{\beta}(x) = h(x) - \beta(c(x) - \mu_c)$$

$$\text{Var}[h_{\beta^*}(X)] = \text{Var}[h(X)] (1 - \rho^2),$$

$$\text{where } \rho = \frac{\text{Cov}(h(X), c(X))}{\sqrt{\text{Var}(h(X)) \text{Var}(c(X))}}.$$

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$$\text{Var}(h_{\beta}) = \text{Var}(h - \beta(c - \mu_c)) = \text{Var}(h) + \beta^2 \text{Var}(c) - 2\beta \text{Cov}(h, c).$$

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Plugging in β^* gives

$$\text{Var}(h_{\beta^*}) = \text{Var}(h) - \frac{\text{Cov}(h, c)^2}{\text{Var}(c)} = \text{Var}(h) (1 - \rho^2).$$

Since $0 \leq \rho^2 \leq 1$, we have $\text{Var}(h_{\beta^*}) \leq \text{Var}(h)$, with **strict** reduction when $|\rho| > 0$.

$$h_{\beta}(x) = h(x) - \beta(c(x) - \mu_c)$$

$$\text{Var}(h_{\beta}(x))$$

$$\text{Var}(h(x))$$

$$\Rightarrow \text{Var}(h_{\beta^*}) = \text{Var}(h) (1 - \rho^2)$$

$$\Rightarrow \rho_1 = 0.9, \rho_2 = 0.5$$

$$\rho$$